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COURSE NAME : ANALOG AND DIGITAL COMMUNICATION (ECA0802)

EXPERIMENT NO: 3A

**GENERATION OF PHASE MODULATED (PM) SIGNALS AND CALCULATION OF PHASE DEVIATION
USING MATLAB SIMULATION**

OBJECTIVES

- i) To generate a Phase Modulated (PM) signal in MATLAB using specified carrier and message signal parameters.
- ii) To observe and analyze the time-domain characteristics of the message signal, carrier signal, and phase modulated signal through MATLAB simulation.
- iii) To calculate the phase deviation of the PM signal using the phase sensitivity constant and message signal amplitude.
- iv) To determine the modulation index of the PM signal and examine its relationship with phase deviation.
- v) To investigate the effect of varying message signal amplitude on the phase deviation and modulation characteristics of the PM signal.
- vi) To compare PM waveforms obtained for different phase deviation values and analyze their modulation behavior using MATLAB.

PROGRAM (MATLAB CODE)

```
clc; clear;

close all;

% Parameters

Am = 1;    % Message Amplitude (V)
fm = 100;  % Message Frequency (Hz)
Ac = 5;    % Carrier Amplitude (V) fc =
1000;      % Carrier Frequency (Hz) fs =
20*fc;     % Sampling Frequency (Hz) kp
= pi/2;    % Phase Sensitivity (rad/V)

% Time Vector

t = 0:1/fs:4/fm;
```

```

% Message Signal m = Am*sin(2*pi*fm*t); % Carrier
Signal c = Ac*cos(2*pi*fc*t); % Phase Modulated
Signal pm = Ac*cos(2*pi*fc*t + kp*m); % Phase
Deviation phase_dev = kp*Am; % PM Modulation
Index mp = phase_dev; % Display Results
fprintf('Message Amplitude = %.2f V\n',Am);
fprintf('Message Frequency = %.0f Hz\n',fm);
fprintf('Carrier Amplitude = %.2f V\n',Ac);
fprintf('Carrier Frequency = %.0f Hz\n',fc);
fprintf('Phase Sensitivity = %.2f rad/V\n',kp);
fprintf('Phase Deviation = %.2f radians\n',phase_dev);
fprintf('PM Modulation Index = %.2f\n',mp);
% Plotting figure('Name','Phase Modulation and Phase
Deviation','NumberTitle','off'); subplot(2,2,1); plot(t,m,'LineWidth',1.5);
title('Message Signal'); xlabel('Time (s)'); ylabel('Amplitude'); grid on;
subplot(2,2,2); plot(t,c,'LineWidth',1.5); title('Carrier Signal'); xlabel('Time
(s)'); ylabel('Amplitude');
grid on; subplot(2,2,3);
plot(t,pm,'LineWidth',1.5);
title('Phase Modulated Signal');
xlabel('Time (s)');
ylabel('Amplitude'); grid on;
subplot(2,2,4); bar(phase_dev);
title('Phase Deviation');
ylabel('Radians');
set(gca,'XTickLabel',{'PM Signal'});
grid on;

```

PROCEDURE

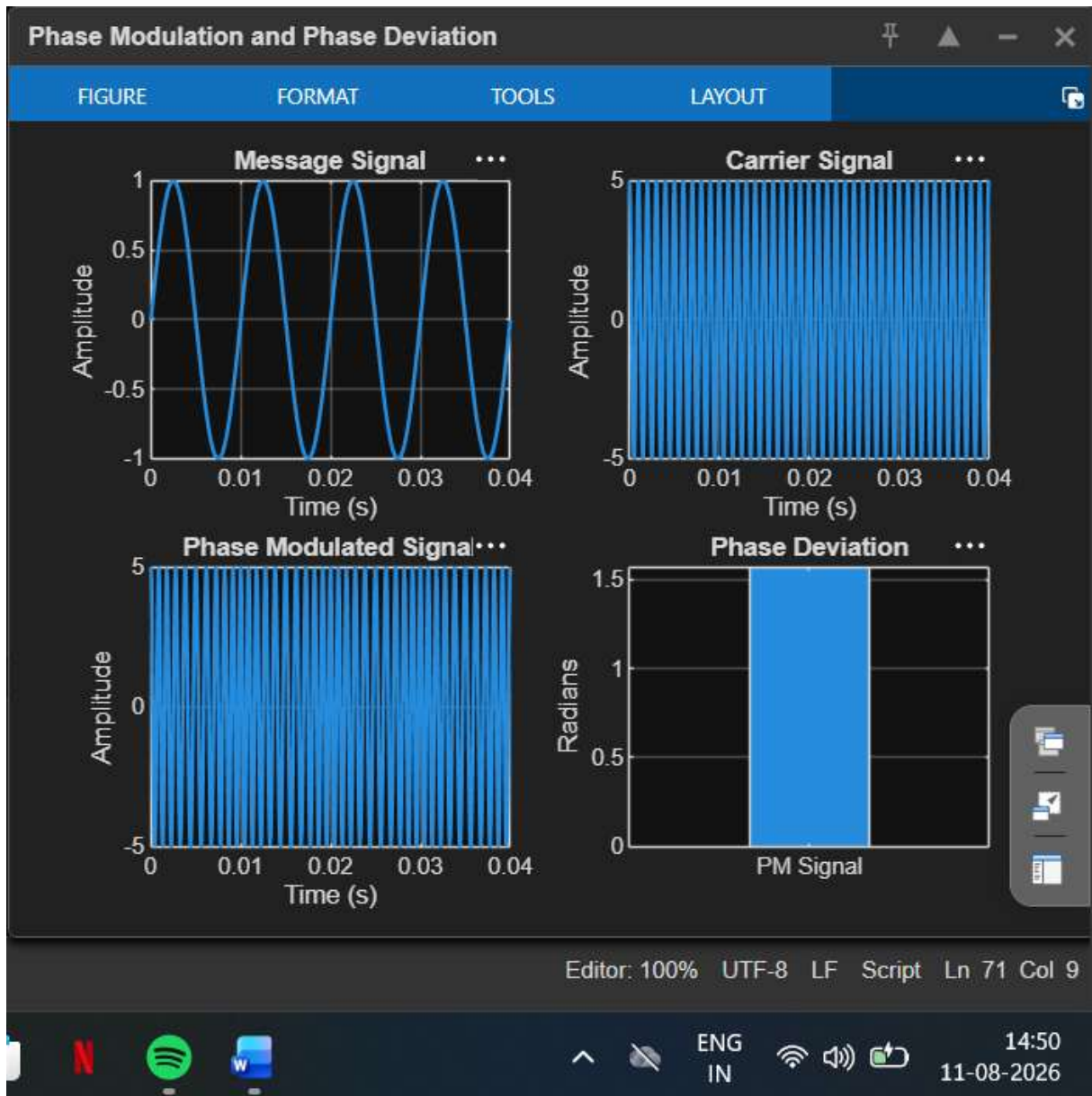
1. Open MATLAB and create a new script file.

2. Initialize the MATLAB workspace using: **clc; clear; close all;**
3. Define the message signal amplitude (A_m), message frequency (f_m), carrier amplitude (A_c), carrier frequency (f_c), sampling frequency (f_s), and phase sensitivity constant (k_p).
4. Generate the time vector using the sampling frequency.
5. Generate the message signal using a sinusoidal waveform.
6. Generate the carrier signal using a sinusoidal waveform.
7. Generate the Phase Modulated (PM) signal using:
$$s(t) = A_c \cos(2\pi f_c t + k_p m(t))$$
8. Plot the message, carrier, and PM signals.
9. Observe phase variations with respect to the message signal.
10. Calculate the phase deviation using:
$$\Delta \theta = k_p \cdot A_m$$

Determine the PM modulation index.

12. Display results using **fprintf()**.
 13. Repeat for different amplitudes and observe changes.
 14. Compare PM waveforms for different deviations.
 15. Record observations and verify relationships.
 16. Analyze and validate PM characteristics.
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OUTPUT



RESULT

The Phase Modulated (PM) signal was successfully generated and analyzed using MATLAB. The phase deviation was calculated from the phase sensitivity constant and message amplitude.

- i) Message Frequency = 100 Hz
- ii) Carrier Frequency = 1000 Hz
- iii) Phase Sensitivity = 1.57 rad/V
- iv) Phase Deviation = 1.57 radians
- v) PM Modulation Index = 1.57
- vi) The PM waveform exhibited phase variations proportional to the message signal amplitude, verifying the characteristics of Phase Modulation.